

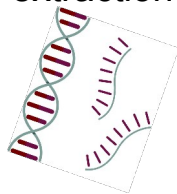
RNA Library Prep for ONT metagenomics

mNGS pipeline

Study
design



Nucleic
acid
extraction



Library prep



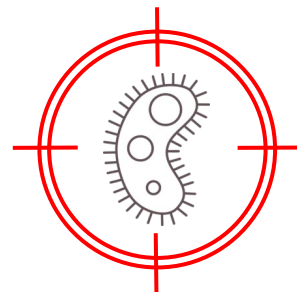
Sequencing



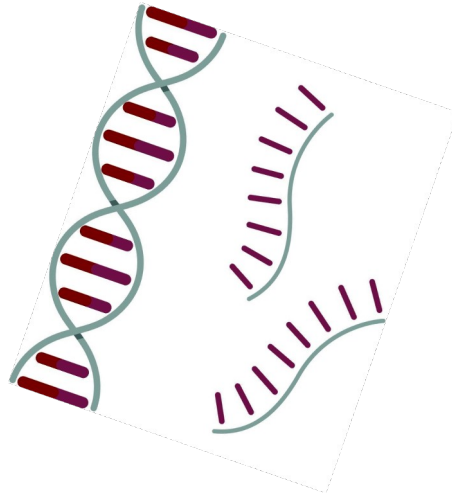
Data analysis

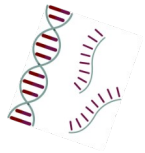


Pathogen ID



Nucleic acid extraction





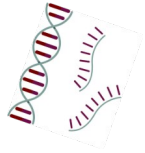
xNA extraction

Key points:

✓ Minimize cross-contamination

✓ Preserve sample quality

- Before extraction:
 - Spin down to remove host cells (extract supernatant)
- Choose type:
 - DNA:
 - Detect bacteria, parasites, fungi, DNA viruses
 - RNA:
 - All the above + RNA viruses (!)
 - Requires optimal sample quality, degrades very fast
 - Keep cold at all times!
 - TNA (total nucleic acid):
 - DNA + RNA
 - Requires DNase I step for RNA-mNGS
- Column vs bead-based extraction



xNA extraction

Zymo DNA+RNA extraction kits



Quick-DNA/RNA Microprep Plus

50 preps for \$399

Column-based

Extra reagents to buy:

- ✓ Molecular grade ethanol (200 proof) for washes
- ✓ Isopropanol to add to concentrated wash buffers
- ✓ DNase I included in regular quick-DNA/RNA kits, but not in Pathogen kits
 - Available separately from Qiagen: RNase-Free DNase Set, \$512 for 250 samples



Quick-DNA/RNA Miniprep Plus

50 preps for \$399

Column-based



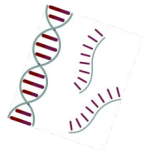
Quick-DNA/RNA Pathogen Magbead

96 preps for \$293

Bead-based

Requires extra:

- ✓ Magnet stand (CZB Engineers)
- ✓ β -mercaptoethanol



xNA extraction

Key point:

✓ xNA quality is crucial

- Quality control:

- Quantity (ng/μl):

- Qubit: RNA or DNA High Sensitivity (HS) Assay Kit (ThermoFisher)

- Quality (length):

- Capillary electrophoresis

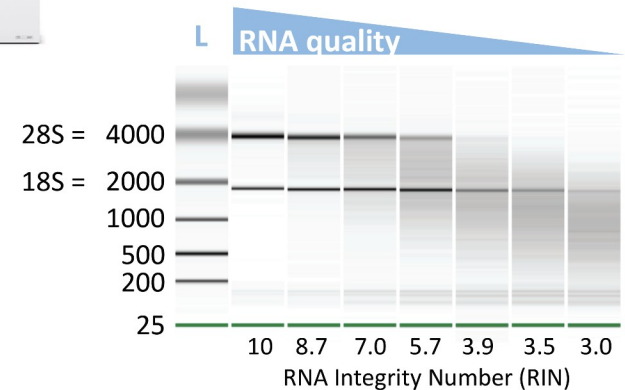
- Tapestation (Agilent)

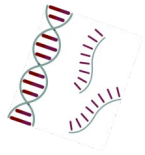
- High Sensitivity DNA or RNA ScreenTape

- Gel electrophoresis

- Regular agarose gel for DNA

- TBE-urea gel for RNA

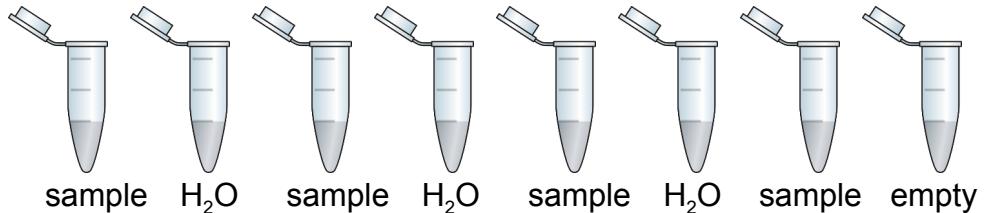




xNA extraction

Keep in mind:

- Magnetic stands for 1.5 ml tubes usually accommodate only 8 tubes at a time
- Cross-contamination should be avoided as much as possible!!
 - Process samples with high pathogen content separately from samples with low pathogen content (if you can)
 - Use water controls to estimate the level of cross-contamination
 - Space out samples with water controls, and/or empty spots in the magnetic rack
 - E.g. for 4 samples:



Next step: Library prep

Considerations when using Nanopore (ONT) sequencing

Sample integrity

- Important to ensure high quality nucleic acid with little to no degradation
- Optimize sample handling and storage

Library preparation kit ([link](#) to available kits)

- DNA or RNA?
- PCR or PCR-free?
- Multiplexing or single sample?

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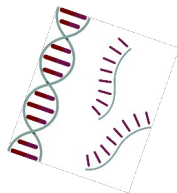
Kit used for ONT RNA mNGS SMART-9N:
Rapid PCR barcoding kit (SQK-RPB114.24)

ONT mNGS pipeline

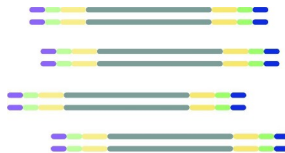
Study
design



xNA
extraction



Library prep



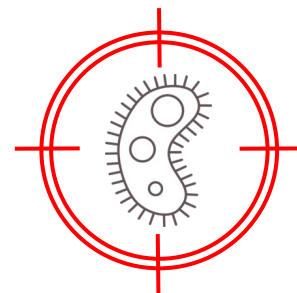
Sequencing



Data analysis



Pathogen ID



RNA mNGS using Nanopore

For RNA mNGS using Nanopore, we use the SMART-9N protocol with minor modifications

- Increase concentration of oligos for reverse transcription steps
- Adjust annealing temperature for PCR

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METHOD ARTICLE 

REVISED Rapid viral metagenomics using SMART-9N amplification and nanopore sequencing

[version 2; peer review: 2 approved]

Ingra M. Claro , Mariana S. Ramundo , Thais M. Coletti , Camila A. M. da Silva , Ian N. Valenca , Darlan S. Candido, Flavia C. S. Sales , Erika R. Manuli, Jaqueline G. de Jesus , Anderson de Paula, Alvina Clara Felix, Pamela dos Santos Andrade, Mariana C. Pinho, William M. Souza , Mariene R. Amorim, José Luiz Proenca-Modena, Esper G. Kallas, José Eduardo Levi, Nuno Rodrigues Faria , Ester C. Sabino , Nicholas J. Loman  , Joshua Quick 

<https://doi.org/10.12688/wellcomeopenres.17170.2>

← → ↻ 

 Josh Quick / Publications / Viral metagenomics using SMART-9n amplification and nanopore sequencing Q Search

Oct 03, 2019

 **Viral metagenomics using SMART-9n amplification and nanopore sequencing**

DOI
[dx.doi.org/10.17504/protocols.io.7w5hpg6](https://doi.org/10.17504/protocols.io.7w5hpg6)

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ARTIC

 **Josh Quick**
University of Birmingham   

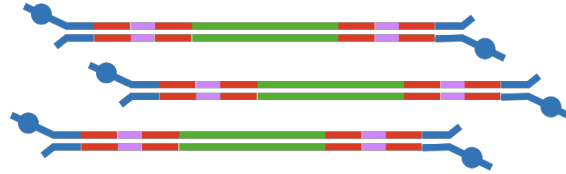
[dx.doi.org/10.17504/protocols.io.7w5hpg6](https://doi.org/10.17504/protocols.io.7w5hpg6)

Library prep: RNA to sequencing library

- Sequencing library = Pool of DNA inserts with sequencing adapters attached

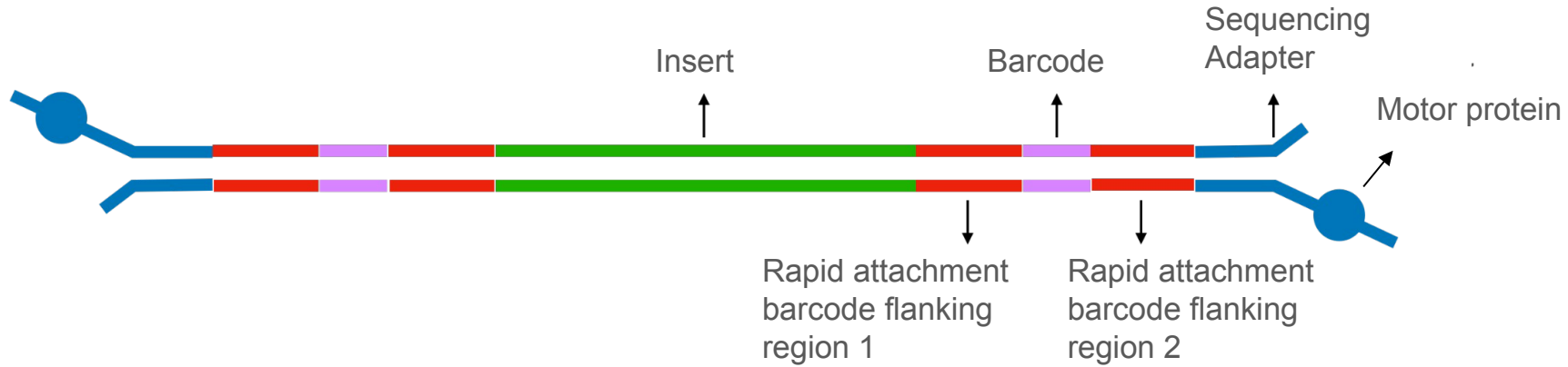


Extracted
RNA

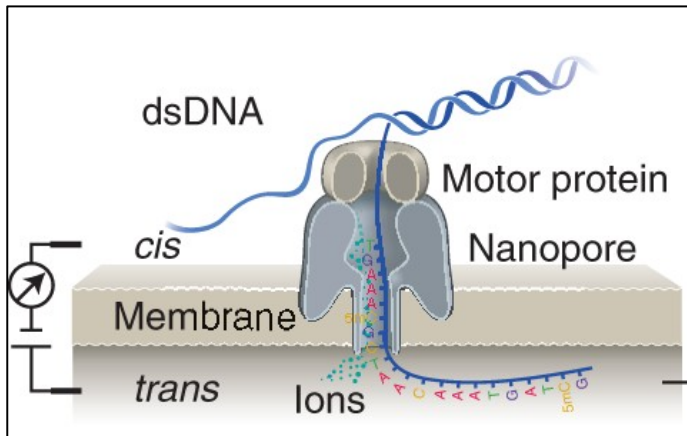
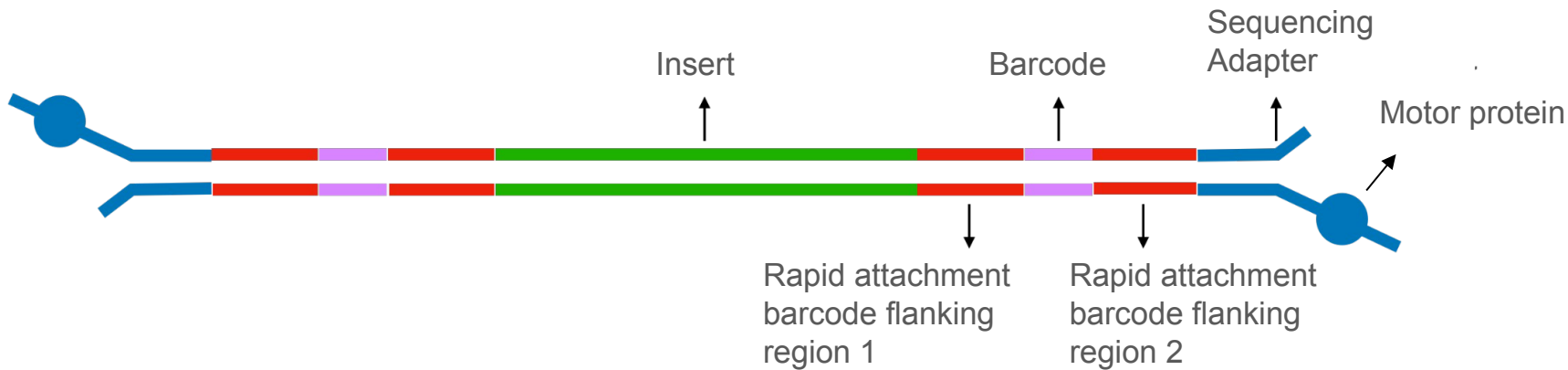


Sequencing
library

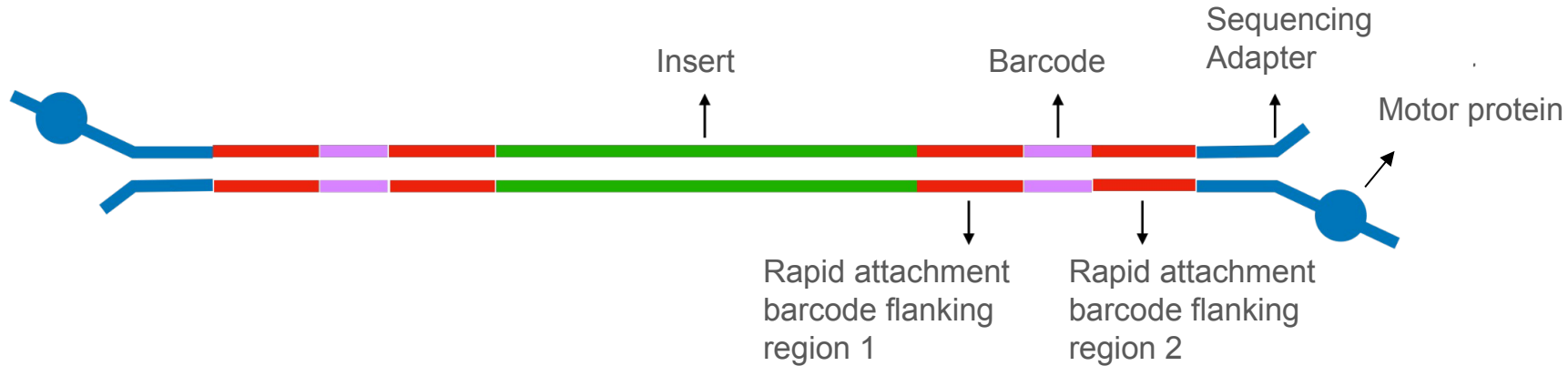
Anatomy of a completed ONT library



Anatomy of a completed ONT library



Anatomy of a completed ONT library

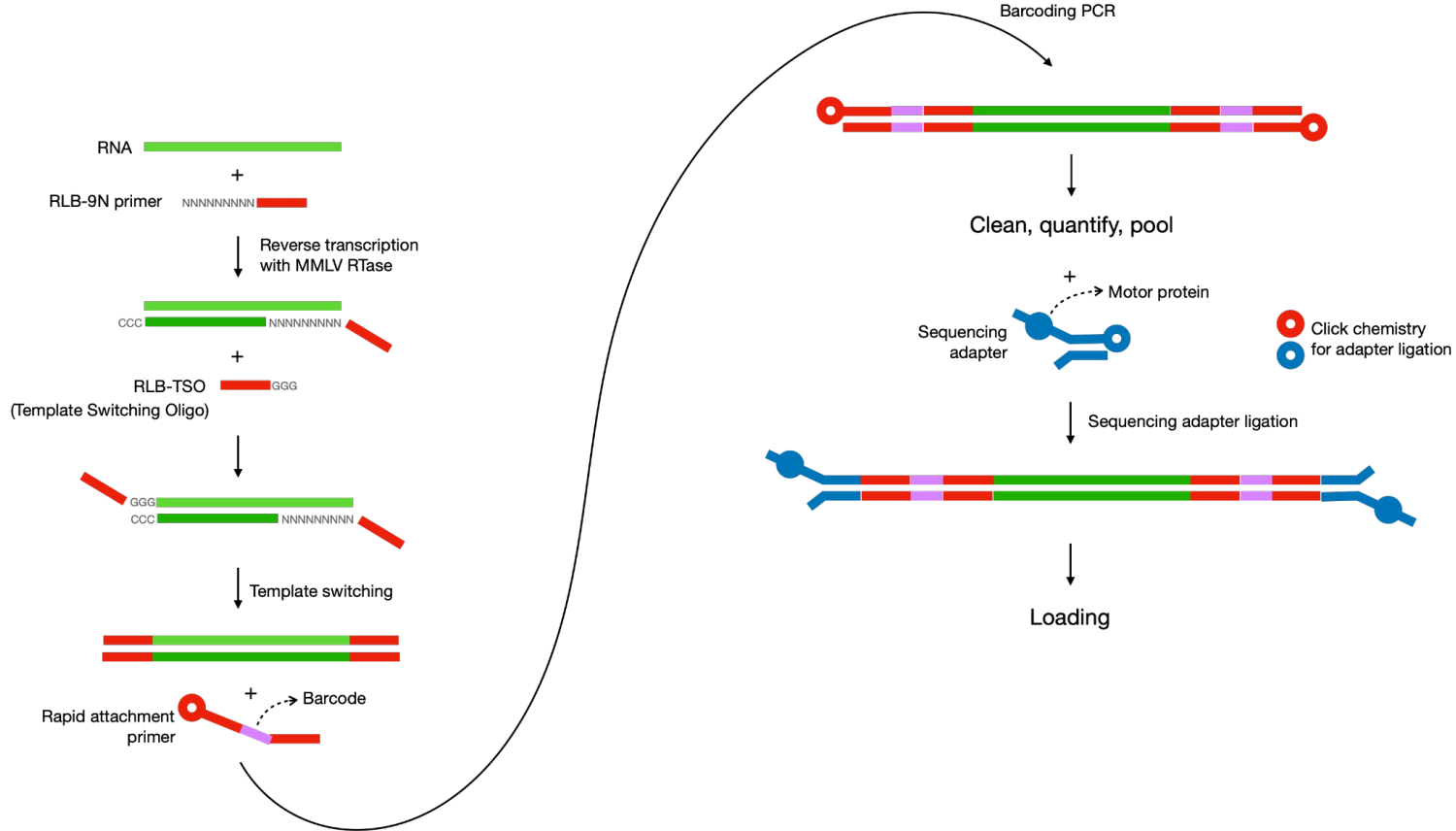


1. **Insert:** cDNA target that will be sequenced
2. **Rapid attachment barcode flanking region 1:** adapter region allows barcode primer to bind for PCR priming; added during cDNA synthesis
3. **Barcode:** DNA barcode specific to each sample; added during PCR amplification
4. **Rapid attachment barcode flanking region 2:** includes special molecule for sequencing adapter ligation; included in barcode primer, added during PCR amplification
5. **Sequencing adapter:** allows binding and movement of DNA through flow cell sequencing pores; includes motor protein; added during sequencing adapter ligation

Steps in RNA mNGS library prep for ONT (SMART-9N)

- Annealing
- Reverse transcription + template switching
- Barcoding PCR
- Clean-up, quantification, and pooling
- Sequencing adapter ligation

RNA mNGS library prep for ONT (SMART-9N)



RNA mNGS library prep for ONT (SMART-9N)

STEP 0. Starting material: RNA or TNA

(pre-PCR lab)

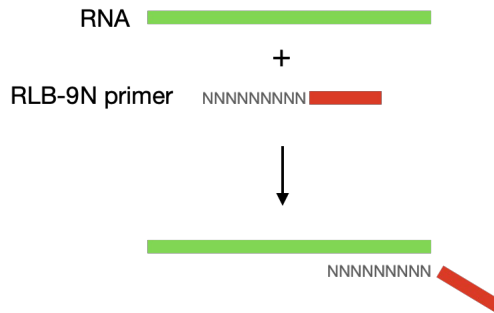
- RNA only: move on to annealing and reverse transcription
- TNA (total nucleic acid): need to remove DNA
 1. Aliquot sample
 2. DNase treatment
 3. SPRI bead cleanup (2x)
 4. Qubit quantification

RNA mNGS library prep for ONT (SMART-9N)

STEP 1. Annealing

(pre-PCR lab)

- RLB-9N primer: ONT adaptor with random nonamer
- ONT adaptor sequence corresponds to the transposase adapter used in Rapid PCR Barcoding kit

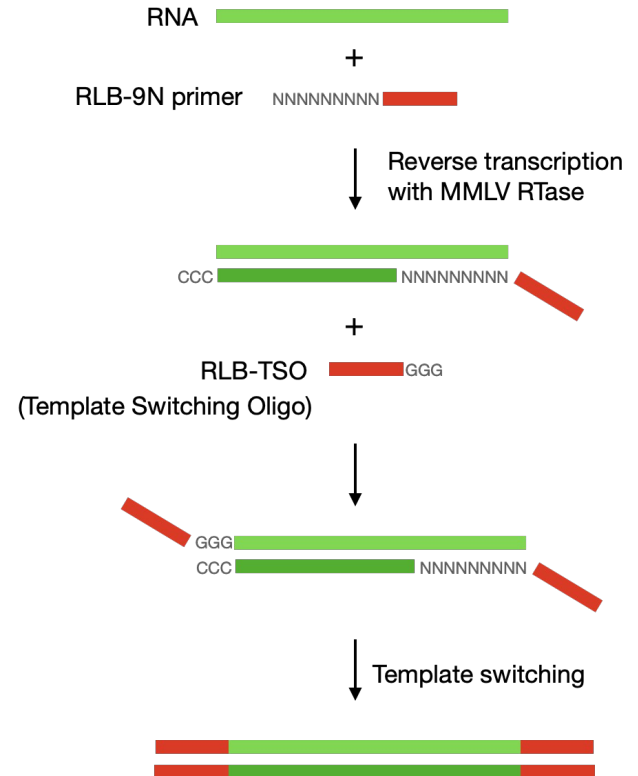


RNA mNGS library prep for ONT (SMART-9N)

STEP 2. Reverse transcription + template switching

switching (pre-PCR lab)

- RNase OUT + DTT → degrade any RNases present
- Special RTase is used that adds CCC at the end (MMLV)
- RLB-TSO: Template Switching Oligo → anneals to CCC and generates the template switched oligo

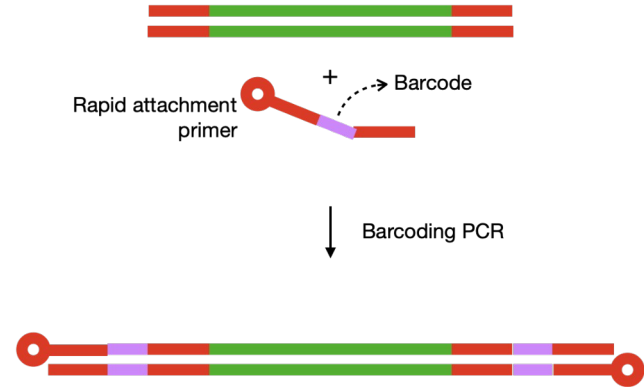


RNA mNGS library prep for ONT (SMART-9N)

STEP 3. Barcoding PCR

(prep pre-PCR lab, amplify in post-PCR)

- Barcoding primers: single barcode, same primer amplifies both ends of each sample
- Barcoding primers have a modified end to facilitate sequencing adapter ligation
- ONT calls these “single rapid attachment primers” and they come with the ONT kit
- PCR conditions:
 - Number of PCR cycles depends on the input; for low input, perform 25 cycles
 - Long extension time used since samples are unfragmented

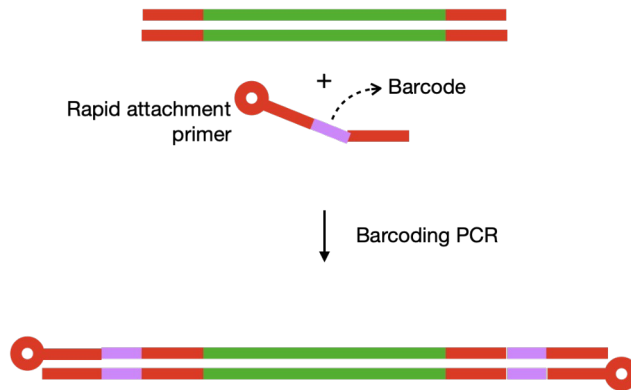


RNA mNGS library prep for ONT (SMART-9N)

STEP 4. Clean-up, QC, and pooling

(post-PCR lab)

- Perform 0.6x SPRI clean-up of libraries after barcoding PCR
- Do QC (quantity and quality)
 - If running Tapestation, use the genomic screentape
 - Ideal size range should be ~1000-7000 bp
- Pooling
 - The goal is to have 100-150 fmol of library in 11μL
 - (This corresponds to a concentration of 10-15 nM)
 - Pooling can be done in nuclease-free water or EB as the kit says (anecdotally, water works better)

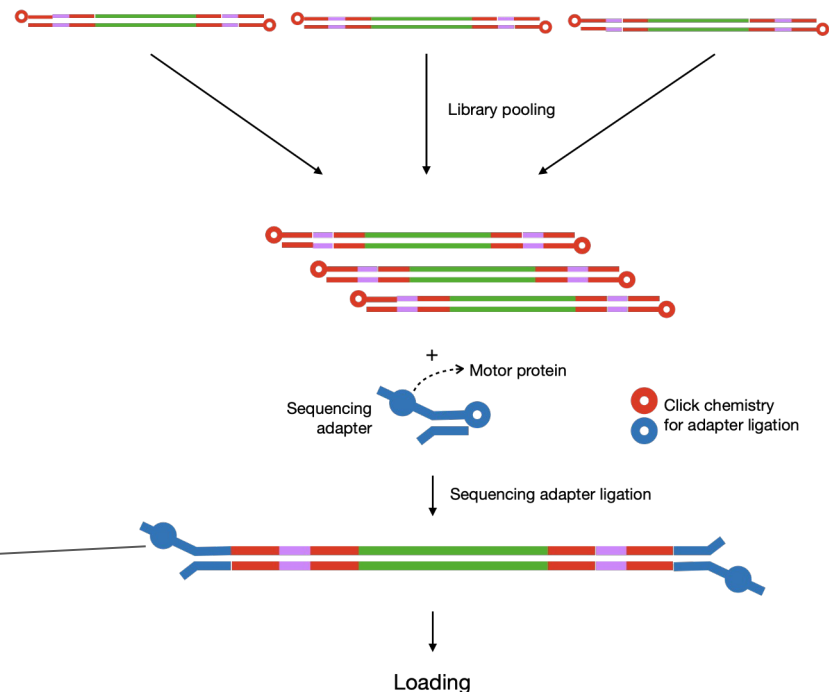
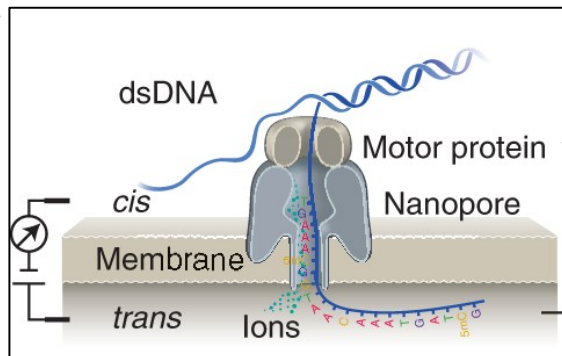


RNA mNGS library prep for ONT (SMART-9N)

STEP 5. Sequencing adapter ligation

(post-PCR lab, immediately before sequencing)

- ONT sequencing adapters have a motor protein attached to it that helps guide the DNA through the nanopores
- Ligation of adapter occurs in 5 min using a proprietary



Resources

RNA and DNA mNGS for ONT:

[CZ Biohub Protocols: Nanopore Metagenomics \(mNGS\)](#)

Next step:
Library QC